Topic 9: Google Sheets Integration

Step 1: Create a New Workflow

1. Open your browser and log into your n8n workspace.
2. In the left navigation, click on **Workflows**.
3. Click the **Add workflow** button in the top-right corner.
4. Rename your workflow in the top-left corner to: `Topic 9 Google Sheets Integration Workflow`.
5. Click the orange **Save** button.

Step 2: Configure the Mock Input Data Node

We will configure mock data to represent a new customer signing up.

1. Click the **Add first step** button in the center of the canvas grid.
2. Search for and select the **When clicking "Test workflow"** node (Manual Trigger).
3. Click the output connector (+ icon) on the right edge of the **Manual Trigger** node.
4. Search for and select the **Edit Fields (Set)** node.
5. Rename this node to `Mock Customer Data`.
6. Double-click the node to open the configuration. Under **Fields to Set**, add the following two string fields:
    a. **Field 1 (String):**
        i. Name: `customerName`
        ii. Value: `Charlie Brown`
    b. **Field 2 (String):**
        i. Name: `customerEmail`
        ii. Value: charlie@example.com
7. Close the configuration panel. Hover over the `Mock Customer Data` node and click the small play icon (**Test step**) to load the payload.

Step 3: Prepare Your Google Sheet

1. Open Google Sheets in your browser and create a new spreadsheet.
2. Rename the spreadsheet to `N8N Test Spreadsheet`.
3. In row 1 (the header row), type the following column headers exactly:
    a. Cell A1: `Name`
    b. Cell B1: `Email`
    c. Cell C1: `Status`
4. Copy your **Spreadsheet ID** from your browser's URL bar (it is the long string of letters and numbers between `/d/` and `/edit` in the URL).

5. Ensure your spreadsheet is shared as **Editor** with the Service Account email configured in your n8n instance (if using Service Account authentication).

Step 4: Add the Google Sheets Append Row Node

1. Click the output connector (+ icon) on the right edge of the `Mock Customer Data` node.
2. Search for and select the **Google Sheets** node.
3. Rename the node to `Append Customer Row`.
4. Double-click the node to configure it:
    a. **Credential Type:** Select your configured Google Sheets API credentials.
    b. **Resource:** Select `Row`.
    c. **Operation:** Select `Append` (or `Append Row`).
    d. **Document:** Paste your copied Spreadsheet ID (or search for `N8N Test Spreadsheet`).
    e. **Sheet:** Select `Sheet1` (or whichever tab name you used).
    f. **Fields to Map:** Under columns mapping, match the sheet columns:
        i. **Name:** `{{ $json.customerName }}` (resolves dynamically)
        ii. **Email:** `{{ $json.customerEmail }}` (resolves dynamically)
        iii. **Status:** Type the static string `Pending`.
5. Close the configuration panel. Hover over the node and click the **Test step** icon to append the row. Open your Google Sheet to verify the new row has been added.

Step 5: Add the Google Sheets Read Node

1. Click the output connector (+ icon) on the right edge of the `Append Customer Row` node.
2. Search for and select another **Google Sheets** node.
3. Rename the node to `Read Customers`.
4. Double-click the node to configure it:
    a. **Credential Type:** Select your Google Sheets credentials.
    b. **Resource:** Select `Row`.
    c. **Operation:** Select `Read` (or `Read Rows`).
    d. **Document:** Select or paste your Spreadsheet ID.
    e. **Sheet:** Select `Sheet1`.
5. Close the configuration panel. Run **Test step** to verify that it reads the appended rows and returns them in JSON format.

Step 6: Add the Google Sheets Update Row Node

1. Click the output connector (+ icon) on the right edge of the `Read Customers` node.
2. Search for and select a third **Google Sheets** node.
3. Rename the node to `Update Customer Status`.
4. Double-click the node to configure it:
    a. **Credential Type:** Select your Google Sheets credentials.
    b. **Resource:** Select `Row`.
    c. **Operation:** Select `Update` (or `Update Row`).
    d. **Document:** Select or paste your Spreadsheet ID.
    e. **Sheet:** Select `Sheet1`.
    f. **Lookup Column:** Select `Email`.
    g. **Lookup Value:** Type the expression `{{ $json.customerEmail }}`.
    h. **Fields to Update:** Select the `Status` column and set its value to `Verified`.
5. Close the configuration panel.

Step 7: Run and Verify the Entire Workflow

1. Click the large orange **Test workflow** button at the bottom-center of the canvas.
2. Once all nodes turn green, inspect your Google Spreadsheet.
3. Verify that the row containing charlie@example.com has had its `Status` cell automatically updated from `Pending` to `Verified`.
4. Click **Save** in the top-right corner to save your workflow.